

Experiment- 3

Half Deflection

AIM:

To determine the resistance of a galvanometer by the half-deflection method and to find its figure of merit.

APPARATUS:

A galvanometer, a voltmeter, a battery or battery eliminator, two (10000 Ω and 200 Ω) resistance boxes, two one-way keys, a rheostat, a screw gauge, a metre scale, an ammeter of given range, connecting wires, and a piece of sandpaper.

THEORY:

The resistance of the given galvanometer as found by the half-deflection method

$$G = \frac{RS}{R - S}$$

Where R is the resistance connected in series with the galvanometer, and S is the shunt resistance.

Figure of merit of the galvanometer:

$$k = \frac{1}{\theta} \frac{E}{R+G}$$

where E is the emf of the battery

PROCEDURE:

1. Make the connections accordingly, as shown in the circuit diagram.
2. See that all plugs of the resistance boxes are tight.
3. Take out the high resistance (2000 Ω) from the resistance box R and insert the key K_1 only.
4. Adjust the value of R so that the deflection is maximum, even in number and within the scale.
5. Note the deflection and let it be θ .
6. Insert the key K_2 also and without changing the values of R, adjust the value of S, such that deflection in the galvanometer reduces to exactly half the value obtained in step 5, i.e; $\frac{\theta}{2}$
7. Note the value of resistance S
8. Repeat steps 4 to 7 three times, taking out different values of R and adjusting S every time.

RESULT:

The resistance of the given galvanometer =

Figure of merit of the galvanometer=

Precautions

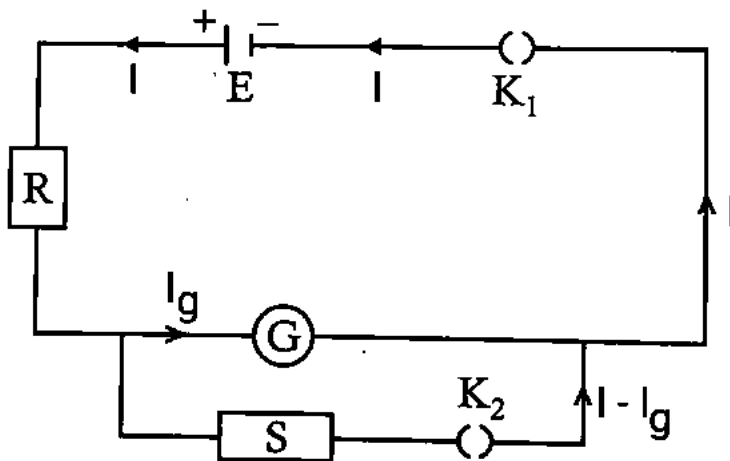
- i. All the plugs in the resistance box should be tight.
- ii. The emf of the cell or battery should be constant.

Sources of Error

- i. The screws of the instruments may be loose.
- ii. The plugs of resistance boxes may not be clean.

Diagram and observation table on the left side of the page

Circuit Diagram:



Observation table

S.No.	Reading R(Ω)	Deflection in the galvanometer (θ)	Shunt resistance S (Ω)	Half deflection($\frac{\theta}{2}$)	Galvanometer resistance $G = \frac{RS}{R-S}$ (Ω)
1					
2					
3					
4					

Calculations:

Mean G=

Figure of merit, $k = \frac{1}{\theta} \frac{E}{R+G}$